

DEFENCE MEDICAL COLLEGE

Final Examination in Histology and Embryology for Year I Medical Students

Time allowed: 3:00 hours.

Date: Feb 18, 2013

Name: _____

I. EMBRYOLOGY

_____ 26. Concerning **gametogenesis**, which statement is **true**?

- a) All oogonia become primary oocytes during ovulation.
- b) The difference in sex chromosome complement of oocytes forms the basis of primary sex determination.
- c) Secondary oocyte complete its second meiotic division to become a mature ovum at fertilization.
- d) Primary oocytes undergo cytodifferentiation to form three polar bodies prior to ovulation.
- e) Primary spermatocytes begin the first meiotic division before birth, but completion of prophase does not occur until adolescence.

_____ 27. Which statement is **true** about **female reproductive system**?

- a) the usual site of fertilization is the ampulla of the uterine tube.
- b) if an oocyte is fertilized, the corpus luteum involutes and degenerates.
- c) the endometrium has a wall made up of a very thick smooth muscle layer.
- d) the perimetrium is a trumpet-shaped opening towards the peritoneal cavity.
- e) the vaginal glands are sloughed off and discarded with each menstrual flow during pregnancy.

_____ 28. With respect to the **zona pellucida**, which statement is **false**?

- a. is composed of sulfated glycoproteins.
- b. prevents polyspermy
- c. prevents premature implantation
- d. functions as a porous filter
- e. arises by delamination from the embryoblast.

_____ 29. Before a **spermatozoan fertilizes an ovum**:

- a) its glycoprotein coat must be removed from its head.
- b) the zona pellucida must be shed.
- c) its acrosomal membrane must be impermeable.
- d) its tail piece should be lost.
- e) it should be capacitated in the epididymidis.

_____ 30. With regard to **fertilization**, which statement is **false**? **Fertilization:**

- a) results in variation of human species through mingling of maternal and paternal chromosomes.
- b) causes metabolic activation of the ovum and initiates cleavage of the zygote.
- c) restores the normal diploid number of chromosomes in the zygote.
- d) determines chromosomal sex of the embryo.
- e) None of the above.

_____ 31. During the **first week of development**,

- _____32. Which of the following is/are **not** a derivative(s) of the **neural crest**?

- d) dizygotic twins always have two amnions and two chorions. e) monozygotic twins are of the same sex.

_____ **40.** Concerning the **development of the urinary System**, which statement is **false**?

- a) Urinary system begins to develop before the genital system
- b) The metanephric diverticulum is the primordium of the ureter.
- c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- d) Differentiation of the nephrons depends upon induction by the collecting tubules.
- e) The kidneys attain their definitive adult position toward the end of the fetal period.

_____ **41.** Concerning the **development of the urinary bladder and urethra**, which statement is **false**?

- a) The female urethra is derived from endoderm of the urogenital sinus.
- b) In adults, the urachus is represented by the median umbilical ligament.
- c) The urinary bladder becomes a pelvic organ during the 9th week of development.
- d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- e) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

_____ **42.** During the **development of the Urinary System**, which statement is **true**?

- a) Bilateral renal agenesis is associated with oligohydramnios.
- b) The podocytes of Bowman's capsule are derived from endoderm.
- c) Connecting tubule is the adult derivative of the ureteric bud.
- d) The kidneys attain their definitive adult position postnatally. e) Urine production begins shortly after birth.

_____ **43.** With regard to the **development of the endocrine system**, which of the following statements is **false**?

- a) the pituitary gland is ectodermal in origin.
- b) the thyroid gland is the first endocrine gland to develop in the embryo.
- c) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.
- d) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
- e) the oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.

_____ **44.** With regard to the **development of gonads**, which statement is **false**?

- a) The surface epithelium of the female gonad develop into oogonia.
- b) Sex of an embryo is determined genetically at the time of fertilization.
- c) Germ cells do not appear in the genital ridges until the 6th week of development.
- d) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
- e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ **45.** With regard to the **vestigial structures derived from embryonic genital system**, which statement is **false**?

- a) A structure found within the adult female pelvis formed from the gubernaculum is median umbilical ligament
- b) The cranial end of mesonephric duct in males may persist as paradidymis.
- c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis.
- d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____46. During the development of the **female reproductive system**, which statement is **false**?

- a) The phallus forms the mons pubis.
- b) The labioscrotal swellings form the labia majora.
- c) The urogenital folds remain unfused and form the labia minora.
- d) The cranial portion of the paramesonephric ducts form the uterine tubes.
- e) During development of the ovaries, the primitive sex (medullary) cords degenerate.

_____47. During the development of the **male reproductive system**, which statement is **false**?

- a) Hypospadias is constantly associated with exstrophy of urinary bladder.
- b) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
- c) Sertoli cells, derived from mesoderm, constitute most of the seminiferous epithelium in the fetal testis.
- d) Interstitial cells (of Leydg) begin to secrete androgenic hormones by the end of the third trimester.
- e) The epididymis, ductus deferens, ejaculatory duct, and seminal vesicle all arise from the paramesonephric ducts.

_____48. During the development of the **reproductive system**, which statement is **true**?

- a) The lower portion of the vagina is derived from hymen.
- b) Gubernaculum does plays a pivotal role in the descent of ovaries.
- c) Dihydrotestosterone inhibits the growth of the paramesonephric duct in males.
- d) In the male, failure of urethral folds to fuse completely results in cryptorchidism.
- e) The most common cause of testicular feminization [Androgen Insensitivity] syndrome is inadequate production of testosterone and müllerian-inhibiting factor.

_____49. With regard to **development of the eye**, which statement is **true**?

- a. The hyaloid canal is found in the vitreous body.
- b. Aqueous humor is produced by the choroid plexus.
- c. Failure of choroid fissure to close results in congenital aphakia.
- d. The optic nerve is a tract of diencephalon that is completely myelinated during the third trimester.
- e. Abnormal development of canal of Schlemm or the iridocorneal filtration angle causes congenital cataract.

_____50. With regard to **development of the ear**, which statement is **true**?

- a. The spiral organ of Corti is derived from neural crest.
- b. The middle ear cavity develops from the otic vesicle.
- c. The utricular portion of the otic vesicle gives rise to cochlear duct.
- d. Perilymph enters the subarachnoid space via the cochlear duct.
- e. None of the above.

f the numbered items or incomplete statements is followed by answers or by completions of the statement.

Select the one lettered answer or completion that is best in each case. (Total : 60 points)

_____1. The ***anterior 2/3rd of the tongue*** develops at which pharyngeal arch level?

- a) 1st
- b) 2nd
- c) 3rd
- d) 4th
- e) 6th

_____2. Which of the following ***salivary glands*** develop from ectoderm?

- a) Parotid b) Submandibular c) Sublingual d) Lingual e) All of the above

_____ 3. All of the following **appendages of the skin** develop from ectoderm, **except**:

- a) Hairs. b) Sweat glands. c) Sebaceous glands. d) Arrector pili muscle. e) Nails.

_____ 4. Concerning the **development of the GIT and associated structures**, which statement is **true**?

- a) Liver develops by fusion of ventral and dorsal hepatic buds.
b) Stomach rotates 270° counterclockwise, similar to the midgut.
c) The loop of the midgut has the superior mesenteric artery in its axis.
d) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.
e) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion develops from the local mesoderm.

_____ 5. With regard to the **congenital anomalies of the kidneys and ureters** which statement is **false**?

- a) Accessory renal arteries are end arteries.
b) Bilateral renal agenesis is associated with oligohydramnios.
c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.
d) Unilateral renal agenesis is more common in males than females. e) None of the above.

_____ 6. Which of the following is not the **adult derivative of the ureteric bud**?

- a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

_____ 7. **Urachal cyst** is a remnant of:

- a) urogenital sinus b) urogenital ridge c) cloaca d) allantois e) mesonephric duct

_____ 8. The **podocytes of Bowman's capsule** are derived from:

- a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

_____ 9. With regard to the development of the **endocrine system**, which of the following statements is **false**?

- a) the oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.
b) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
c) the thyroid gland is the first endocrine gland to develop in the embryo.
d) the pituitary gland is ectodermal in origin.
e) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.

_____ 10. A structure found within the adult **female pelvis formed from the gubernaculum** is

- a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament

_____ 11. During the **development of the kidney systems**, which statement is **true**?

- a) The transitional epithelium lining the urethra is derived from neuroectoderm.
b) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.
d) Urine production begins shortly after birth. e) At birth, the kidneys have a lobulated appearance.

_____ 12. The **labia minora** arise embryologically from:

- 13.** The ***uterine tubes*** of the adult female are derived embryologically from
a) mesonephric duct b) mesonephric tubules c) paramesonephric duct
d) paramesonephric tubules e) uterovaginal primordium

_____ 14. The most common cause of **testicular feminization** [Androgen Insensitivity] **syndrome** is

a) a 46, XO genotype b) a 47, XXY genotype c) lack of androgen receptors

d) congenital adrenal hyperplasia e) inadequate production of testosterone and müllerian-inhibiting factor.

 15. In the male, **failure of urethral folds to fuse completely** results in
a) hypospadias b) epispadias c) cryptorchidism d) hydrocele e) congenital inguinal hernia

16. Intermediate medoderm gives rise to the
a) neural tube b) heart c) kidneys and gonads d) somites e) notochord

_____ 17. Concerning the **development of the Urinary System**, which statement is **false**?

- a) The kidneys attain their definitive adult position postnatally.
- b) Urinary system begins to develop before the genital system.
- c) The metanephric diverticulum is the primordium of the ureter.
- d) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- e) Differentiation of the nephrons depends upon induction by the collecting tubules.

_____ 18. Which of the following is not the adult derivative of the **ureteric bud**?
a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

19. Concerning the **development of the urinary bladder and urethra**, which statement is **false**?

- a) The female urethra is derived from endoderm of the urogenital sinus.
- b) In adults, the urachus is represented by the median umbilical ligament.
- c) The urinary bladder becomes a pelvic organ during the 9th week of development.
- d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- e) Epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

 20. Up to the 5th week of development, the embryonic **skin** is only a single layer of:

a) Ectoderm b) Endoderm c) Paraxial mesoderm
d) Intermediate mesoderm e) Lateral plate mesoderm

_____ 21. During the development of the female reproductive system, which statement is false?

- a) The urogenital folds remain unfused and form the labia minora.
- b) The cranial portion of the paramesonephric ducts form the uterine tubes.
- c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.
- d) The phallus forms the mons pubis.
- e) The labioscrotal swellings form the labia majora.

22. Concerning the **development of the genital system**, one statement is **false**?

a) Bilateral cryptorchidism may result in sterility.

b) Hypospadias is constantly associated with exstrophy of urinary bladder.

- c) In testicular feminization syndrome, uterus and uterine tubes are generally absent.
- d) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora
- e) In epispadias usually the penis is underdeveloped and is curved ventrally, a condition known as chordee.

_____ **23.** With regard to the **development of gonads**, which statement is **false**?

- a) The surface epithelium of the female gonad develop into oogonia.
- b) Sex of an embryo is determined genetically at the time of fertilization.
- c) Germ cells do not appear in the genital ridges until the 6th week of development.
- d) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
- e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ **24.** With regard to the **vestigial structures derived from embryonic genital ducts**, which statement is **false**?

- a) The prostatic utricle is homologous to the vagina.
- b) The cranial end of mesonephric duct in males may persist as paradidymis.
- c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis
- d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ **25.** Concerning the development of the **Integumentary system**, which statement is **true**?

- a) castoff cells from the epidermis mix with secretions of sebaceous glands to form the vernix caseosa.
- b) toenails and fingernails begin to develop at the tips of the digits postnatally.
- c) teeth develop from ectoderm, mesoderm and neural crest cells.
- d) eccrine sweat glands develop from mesenchyme, which is derived from the mesoderm underlying the surface ectoderm.
- e) the epidermis is derived from surface ectoderm.

_____ **26.** During the **development of the digestive system**, which statement is **true**?

- a) omphalocele is a persistence of the herniation of abdominal contents into the proximal part of the umbilical cord.
- b) the superior two-thirds of the adult anal canal are derived from proctodeum.
- c) the midgut may give rise to the descending colon and sigmoid colon.
- d) the dorsal pancreatic bud forms the uncinat process and part of the head of the pancreas.
- e) the anal membrane normally breaks down by the end of the second trimester.

_____ **27.** What germ layer forms the melanocytes?

- a) Endoderm
- b) Neural crest
- c) Paraxial mesoderm
- d) Intermediate mesoderm
- e) Lateral plate mesoderm

_____ **28.** What germ layer is the **dermis** derived from?

- a) Paraxial mesoderm
- b) Intermediate mesoderm
- c) Lateral plate mesoderm
- d) Neural crest
- e) Endoderm

_____ **29.** At birth the infant is covered by a whitish paste (vernix caseosa), which protects the skin against the macerating action of amniotic fluid. Which of the following is the **vernix caseosa** derived from?

- a) Arrector pili muscles
- b) Dermal root sheaths
- c) Hair papilla
- d) Sebaceous glands
- e) Hair blood vessels

- _____ **30.** During a routine examination of a female patient, **accessory nipples** are found in the axilla and on the mid thorax along the embryological mammary ridge. What is the terminal extension of this ridge?
a) Mid thorax b) Diaphragm c) Umbilicus d) Pubic region e) Lower thigh to knee
- _____ **31.** From what layer of the epidermis do **hair follicles** begin to proliferate?
a) Stratum corneum b) Stratum lucidum c) Stratum granulosum
d) Stratum spinosum e) Stratum germinativum
- _____ **32.** This disorder results from **excessive keratinization of the skin**, whereby the skin appears as very dry and scaly. In extreme cases (harlequin fetus), the skin is abnormally thick and cracked.
a) Albinism b) Amelogenesis imperfect c) Leprosy d) Psoriasis e) Ichthyosis
- _____ **33.** A patient presents with **red blotches on their face and neck**. They say they've had these birthmarks as long as they can remember and they were told it has something to do with capillaries. Which of the following malformation does this patient likely have?
a) Albinism b) Angioma c) Ichthyosis d) Collodion e) Thalidomide defect
- _____ **34.** The **midgut** begins with _____ and ends with _____.
a) 1/2 jejunum; 2/3 transverse colon b) 1/2 ileum; 2/3 sigmoid colon c) 1/2 jejunum; 2/3 rectum
d) 1/2 duodenum; 2/3 transverse colon e) 1/2 duodenum; 1/3 sigmoid colon
- _____ **35.** The **middle of the duodenum** would receive blood from:
a) Superior mesenteric artery only b) Inferior mesenteric artery only c) Celiac artery only
d) Celiac and inferior mesenteric artery e) Celiac and superior mesenteric artery
- _____ **36.** What two structures only form on the **ventral side of the bile duct**?
a) Liver; Pancreas b) Liver; Gallbladder c) Pancreas; Gallbladder
d) Heart; Liver e) Heart; Pancreas
- _____ **37.** During week six, the **midgut herniates** through the umbilical ring, rotating _____ degrees counter clockwise. Near week ten, the midgut returns, rotating _____ degrees for an overall rotation of _____ degrees.
a) 180; 180; 0 b) 180; 180; 360 c) 90; 180; 90 d) 90; 180; 270 e) 270; 90; 180
- _____ **38.** A three-week old male infant comes into the Emergency Room for **weight loss and acute dehydration**. The infant has been **projectile vomiting** and the condition is worsening. Which of the following is most likely malformation?
a) Gastric atresia b) Pyloric stenosis c) Annular pancreas
d) Malrotation of the gut e) Intestinal duplication
- _____ **39.** **Meckel's diverticulum** is a congenital connection from the umbilicus via a vitelline ligament to
a) Duodenum b) Jejunum c) Ileum d) Transverse colon e) Sigmoid colon
- _____ **40.** During the **pronephros period** (week 3), the first kidney develops in the _____ region. By week 4 this kidney degenerates.
a) Cervical b) Thoracic c) Lumbar d) Sacral e) Cloaca
- _____ **41.** The **ureteric bud** gives rise to the _____ and the **metanephric mesoderm** gives rise to the _____.
a) Nephrons; Collecting system b) Collecting system; Nephrons c) Bowman's capsule; Ureter

d) Loop of Henle; Renal pelvis

e) Distal/proximal convoluted tubules; Major/minor calyces

_____ **42. The urorectal septum** divides the _____ from the _____.

- a) Anorectal canal; Urogenital sinus b) Urogenital sinus; Mesonephric duct c) Urachus; Allantois
d) Urinary bladder; Allantois e) Mesonephric duct; Ureteric bud

_____ **43. The urachus** forms the:

- a) Allantois b) Median umbilical ligament c) Medial umbilical ligament
d) Lateral umbilical ligament e) Round (umbilical) ligament

_____ **44. The bladder** becomes continuous with the lower end of the mesonephric (Wolffian) duct, which will form the _____ of the bladder.

- a) Urethra b) Prostate c) Trigone d) Posterior wall e) Anterior wall

_____ **45. At birth, a child has large palpable flank masses and abnormal extremities.** The delivery physician is very concerned as she knows this **autosomally-dominant disorder** can lead to renal failure and the child will likely need a kidney transplant to survive. Which of the following disorders does the child likely have?

- a) Polycystic kidney b) Renal agenesis c) Pelvic kidney d) Bifid ureter e) Bifid kidney

_____ **46. This disorder, associated with oligohydramnios, occurs when the ureteric bud fails to reach the metanephric mesoderm.**

- a) Polycystic kidney b) Renal agenesis c) Pelvic kidney
d) Horseshoe kidney e) Bifid ureter

_____ **47. What artery prevents the horseshoe kidney (metanephric mesodermal fusion) from ascending high into the abdomen?**

- a) Celiac trunk b) Superior mesenteric artery c) Inferior mesenteric artery
d) Left renal artery e) Right renal artery

_____ **48. A newborn baby presents with fluid draining from the umbilicus onto the skin.** Testing of the fluid identifies it as urine. What is the most likely diagnosis?

- a) Urachal cyst b) Urachal fistula c) Exstrophy of the bladder d) Bifid ureter e) Horseshoe kidney

_____ **49. A patient is born with karyotype 46, XY but has no problem with production of testosterone or müllerian inhibiting hormone. Their vagina ends in a blind pouch and testes are lodged in the inguinal canal.** Which of the following is the best description of this patient:

- a) Male pseudo-hermaphrodite b) Female pseudo-hermaphrodite c) Androgen insensitivity syndrome
d) Turner's syndrome e) Klinefelter syndrome

_____ **50. During development of the ovaries, the primitive sex (medullary) cords _____.**

- a) Proliferate b) Move laterally c) Move medially d) Differentiate e) Degenerate

_____ **51. The gubernaculum** is attached to the uterus and as the ovaries descend it becomes the _____ of the uterus.

- a) Canal of Nuck b) Ovarian ligament c) Round ligament
d) Ovarian and round ligaments e) Gubernacular ligament

_____ **52. Development of the testis** is induced by the sry-gene, which encodes testis-determining factor

(TDF). TDF causes all of the following **EXCEPT**:

- a) The sex cords continue to proliferate to form the testes.
- b) Leydig cells are secreting testosterone thus have very little SER.
- c) Towards the hilum, the cords grow into a network that will give rise to the rete testes.
- d) During further development, a thick layer (tunica albuginea) separates the cords from the surface epithelium.
- e) Sex cords become seminiferous cords that form seminiferous tubules (Sertoli cells), straight tubules, and rete testes

_____ **53.** Which of the following inhibits the growth of the **paramesonephric duct (males)?**

- a) Testosterone b) Leydig Cells c) Dihydrotestosterone d) Estrogens e) Müllerian substances (MIFs)

_____ **54.** As the **testes descend**, they follow the:

- a) Processus vaginalis b) Parietal layer of tunica vaginalis c) Visceral layer of tunica vaginalis
- d) Caudal genital ligament e) Cryptorchidism

_____ **55.** The **epididymis, ductus deferens, ejaculatory duct, and seminal vesicle** all arise from:

- a) The Müllerian ducts b) The paramesonephric ducts c) The mesonephric ducts
- d) Leydig hormones e) Wolffian hormones

_____ **56.** The **uterine tubes, uterus, cervix, and superior part (1/3) of the vagina** all arise from:

- a) The Wolffian ducts b) The paramesonephric ducts c) The mesonephric ducts
- d) Leydig hormones e) Müllerian hormones

_____ **57.** The **epioophoron and paroophoron**, which are important as cysts, arise from:

- a) The Müllerian ducts b) The paramesonephric ducts c) The mesonephric ducts
- d) Leydig hormones e) Wolffian hormones

_____ **58.** The **lower portion of the vagina** is derived from:

- a) Sinovaginal bulb b) Uteral lumen c) Hymen d) Urogenital sinus e) Vaginal plate

_____ **59.** What structure in the **indifferent stage** (weeks 3-6) becomes the **clitoris**?

- a) Urogenital groove b) Genital swelling c) Urethral fold d) Perineum e) Genital tubercle

_____ **60. Genetic 46,XY** would be:

- a) Turner's syndrome b) Klinefelter syndrome c) True hermaphrodite
- d) Male pseudo-hermaphrodite e) Female pseudo-hermaphrodite

_____ **1.** Most **birth defects** occur during which of the following periods/stages of the intrauterine life?

- a) First 2 weeks b) Embryonic period c) Fetal period d) Last trimester e) Zygote

_____ **2.** Which of the following statements is **not true** regarding the **human placenta**?

- a) Maternal portion is formed by decidua parietalis. b) Fetal portion is formed by chorion frondosum.
- c) Cotyledons are seen from the maternal aspect. d) Fetal portion is smooth.
- e) At full term covers the whole surface area of the endometrium.

- _____ 3. The **central tendon of the diaphragm** develops from:
- a) Dorsal mesentery of the esophagus b) Pleuroperitoneal membranes
c) Pleuropericardial membranes. d) Lateral body wall e) Septum transversum.
- _____ 4. Which of the following is not a component of **vernix caseosa**?
- a) Secretions from sebaceous gland b) Secretions of apocrine sweat glands
c) Fallen epidermal cells d) Fallen hairs e) None of the above
- _____ 5. Structure that is **not** developed in the **third branchial arch** include:
- a) Lesser cornu of hyoid bone b) Body of the hyoid bone
c) Glossopharyngeal nerve d) Common carotid artery e) Stylopharyngeus muscle
- _____ 6. Which of the following structures develop from **endoderm of the second pharyngeal pouch**?
- a) Thymus gland b) Palatine tonsil c) Thyroid gland
d) Superior parathyroid gland e) Parafollicular cells of thyroid gland
- _____ 7. The **anterior 2/3rd of the tongue** develops at which pharyngeal arch level?
- a) 1st b) 2nd c) 3rd d) 4th e) 6th
- _____ 8. Which of the following **salivary glands** develop from ectoderm?
- a) Parotid b) Submandibular c) Sublingual d) Lingual e) All of the above
- _____ 9. *Secretion of surfactant, which is an important substance for survival of premature infants appears during which of the following periods of lung development?*
- a) *Canalicular.* b) *Bronchiolar.* c) *Terminal sac period* d) *Pseudoglandular* e) *Alveolar*
- _____ 10. Regarding the **development of the gut**, which of the following statements is **correct**?
- a) The lumen of the intestine is temporarily blocked by epithelial proliferation.
b) The site of junction of foregut and midgut derivatives is the middle of jejunum.
c) Is formed as the result of absorption of part of the amniotic cavity because of the embryonic foldings.
d) The midgut is temporarily connected to the yolk sac by the allantois.
e) All of the above.
- _____ 11. All of the following **appendages of the skin** develop from ectoderm, except:
- a) Hairs. b) Sweat glands. c) Sebaceous glands. d) Arrector pili muscle. e) Nails.
- _____ 12. With regard to **fertilization**, which statement is **false**?
- a) restores the normal diploid number of chromosomes in the zygote.
b) results in variation of human species through mingling of maternal and paternal chromosomes.
c) causes metabolic activation of the ovum and initiates cleavage of the zygote.
d) determines chromosomal sex of the embryo. e) None of the above.
- _____ 13. During the **first week of development**,
- a) the embryo is a solid ball of blastomeres and is called gastrula.
b) mesoderm is formed which will give rise later to the placental membranes.
c) the blastocyst is superficially implanted in the uterine wall (endometrium).
d) developmental disturbances may give rise to major congenital anomalies.

e) the amniotic cavity is developed within the yolk sac as a space between the cytotrophoblast and the embryoblast.

_____ **14. During the third week of development, which statement is **false**?**

- a) somites, which will give rise to the vertebrae, ribs and axial musculature, are formed.
- b) notochord forms the primitive axis of the embryo around which the axial skeleton will form.
- c) the primitive streak appears as a thickening of the embryonic epiblast at the caudal end of the embryonic disc.
- d) mesoderm exists between the ectoderm and endoderm everywhere.
- e) neural tube is formed.

_____ **15. During the fetal period,**

- a) the spleen is the major site of erythropoiesis during the end of the 9th week.
- b) ossification of the fetal skeleton is active during the 13th -16th week.
- c) the bodies of 20- week fetuses are usually completely covered with lanugo hairs.
- d) by 18 weeks the uterus is formed and canalization of the vagina has begun.
- e) the amount of white fat is about 16% of body weight at 29 weeks.

_____ **16. With regard to the placenta,**

- a) the maternal part of the placenta develops from the chorionic sac.
- b) the fetal part of the placenta is attached to the maternal part of the placenta by the cytotrophoblastic shell.
- c) amniochorionic membrane ruptures during labor.
- d) as pregnancy advances, the placental membrane becomes progressively thinner.
- e) the fetal surface of a freshly delivered placenta has the characteristic cobblestone appearance produced by cotyledons.

_____ **17. Concerning the fetal membranes (developmental adnexa),**

- a) the yolk sac helps control the embryo's body temperature.
- b) primordial germ cells appear in the endodermal lining of the wall of the allantois.
- c) during the 4th week the endoderm of the yolk sac is incorporated into the embryo as the primordial gut.
- d) as the bladder enlarges, the umbilical cord involutes to form urachus.
- e) blood formation occurs in the wall of allantois during the third to fifth weeks.

_____ **18. Concerning multiple pregnancies,**

- a) twins with one amnion, one chorion and one placenta are always monozygotic.
- b) the rate of dizygotic twinning increases with maternal age.
- c) the frequency of monozygotic twinning shows marked racial differences.
- d) dizygotic twins always have two amnions and two chorions.
- e) monozygotic twins are of the same sex.

_____ **19. The neuroepithelial cells of the neural tube gives rise to all of the following cells of the brain except:**

- a) Microglia.
- b) Astrocytes.
- c) Oligodendrocytes.
- d) Purkinje cells of cerebellar cortex.
- e) Ependymal cells.

_____ **20. With regard to birth defects,**

- a) the phenotype of Turner syndrome is female.
- b) thalidomide induces limb defects.
- c) trisomy of the autosomes occurs with increasing frequency as maternal age increases.
- d) deafness has been reported in infants of mothers who have been treated with penicillin during the 2nd trimester.
- e) phenobarbital is considered to be a safe, antiepileptic drug for use during pregnancy.

_____ **21. Concerning the development of the GIT and associated structures:**

- a) The loop of the midgut has the superior mesenteric artery in its axis.
- b) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion develops from the local mesoderm.
- c) Liver develops by fusion of ventral and dorsal hepatic buds.
- d) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.
- e) Stomach rotates 270° counterclockwise, similar to the midgut.

_____ **22. With regard to the development of the musculoskeletal system,**

- a) the muscular system develops from neuroectoderm.
- b) the upper limbs rotate laterally through 90° on their longitudinal axes.
- c) premature closure of the frontal (metopic) suture results in scaphocephaly.
- d) sclerotomes of the somites form the viscerocranium. e) xiphoid process develops during the 6th week.

_____ **23. *Cerebellum* develops from cells of:**

- a) Basal plate of myelencephalon
- b) Alar plate of myelencephalon
- c) Basal plate of metencephalon
- d) Alar plate of myelencephalon
- e) Basal plate of mesencephalon

_____ **24. *Primary brain vesicle* that remains relatively unchanged into the secondary brain vesicle is:**

- a) Rhombencephalon
- b) Prosencephalon
- c) Telencephalon
- d) Mesencephalon
- e) Diencephalon

_____ **25. Which of the following components plays the most active role in *invading the endometrium during blastocyst implantation*?**

- a) epiblast
- b) syncytiotrophoblast
- c) hydroblast
- d) extraembryonic somatic mesoderm
- e) extraembryonic visceral mesoderm

_____ **26. The *prochordal plate* marks the site of the future**

- a) umbilical chord
- b) heart
- c) mouth
- d) anus
- e) nose

_____ **27. Which process establishes the *three definitive germ layers*?**

- a) neurulation
- b) gastrulation
- c) craniocaudal folding
- d) lateral folding
- e) angiogenesis

_____ **28. The *spiral organ of Corti* is derived from**

- a) both ectoderm and mesoderm
- b) neural crest
- c) endoderm
- d) mesoderm
- e) surface ectoderm

_____ **29. The *middle ear cavity***

- a) is of mesodermal origin
- b) develops from the otic vesicle
- c) develops from pharyngeal arch 1
- d) develops from pharyngeal pouch 2
- e) develops from pharyngeal pouch 1

_____ **59. With regard to the congenital anomalies of the kidneys and ureters which statement is false?**

- a) Accessory renal arteries are end arteries.
- b) Bilateral renal agenesis is associated with oligohydramnios.
- c) Incontinence is the common complaint resulting from an ectopic ureteric orifice in both sexes.
- d) Unilateral renal agenesis is more common in males than females.
- e) None of the above.

- _____ **30.** The **saccular portion of the otic vesicle** gives rise to the
a) organ of Corti _____ b) endolymphatic duct c) superior semicircular canal
d) crus commune nonampullare e) lateral semicircular canal
- _____ **31.** **Perilymph** enters the subarachnoid space via the
a) cochlear duct b) ductus reuniens c) perilymphatic duct
d) vestibular aqueduct e) utriculosaccular duct
- _____ **32.** Which of the following structures in the **eye** is derived from **surface ectoderm**?
a) retina b) dilator and sphincter pupillae muscles of iris
c) corneal endothelium d) vitreous body e) lens
- _____ **33.** Which of the following is not the adult derivative of the **ureteric bud**?
a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule
- _____ **34.** **Urachal cyst** is a remnant of
a) urogenital sinus b) urogenital ridge c) cloaca d) allantois e) mesonephric duct
- _____ **35.** The **podocytes of Bowman's capsule** are derived from
a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells
- _____ **36.** A structure found within the adult **female pelvis formed from the gubernaculum** is
a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament
- _____ **37.** After the **sinovaginal bulbs** have proliferated and fused, they form a solid core of endodermal cells called
a) vestibule of the vagina b) uterovaginal primordium c) urogenital sinus
d) vaginal plate e) clitoris
- _____ **38.** The **labia minora** arise embryologically from
a) phallus b) labioscrotal swellings c) sinovaginal bulbs
d) urogenital folds e) paramesonephric duct
- _____ **39.** The **uterine tubes** of the adult female are derived embryologically from
a) mesonephric duct b) mesonephric tubules c) paramesonephric duct
d) paramesonephric tubules e) uterovaginal primordium
- _____ **40.** The most common cause of **testicular feminization syndrome** is
a) a 46, XO genotype b) a 47, XXY genotype
c) lack of androgen receptors d) congenital adrenal hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)
- _____ **41.** In the male, **failure of urethral folds to fuse completely** results in
a) hypospadias b) epispadias c) cryptorchidism d) hydrocele e) congenital inguinal hernia
- _____ **42.** The **Y chromosome** carries a gene on its short arm that codes for

a) testosterone b) MIF c) testes-determining factor (TDF) d) progesterone e) estrogen

_____ **48.** Which of the following best describes the **placental components of dizygotic twins**?

- a) one placenta, two amniotic sacs, one chorion b) one placenta, two amniotic sacs, two chorions
c) two placentas, two amniotic sacs, one chorion d) two placentas, two amniotic sacs, two chorions
e) one placenta, two amniotic sacs, two chorions

_____ **43.** The **notochord** induces the overlying ectoderm to develop into the

- a) skin b) eye c) primitive streak d) primitive node e) neural plate

_____ **44.** A **sacrooccygeal teratoma** is a tumor that arises from remnants of the

- a) neural plate b) cloacal membrane c) posterior neuropore d) primitive streak e) notochord

_____ **45.** **Intermediate medoderm** gives rise to the

- a) neural tube b) heart c) kidneys and gonads d) somites e) notochord

_____ **46.** *The proximal part of the right subclavian artery is derived from aortic arch*

- a) 1 b) 2 c) 3 d) 4 e) 6

_____ **47.** *The portal vein is derived from*

- a) vitelline veins b) *umbilical veins* c) *anterior cardinal veins*
e) *posterior cardinal veins* e) *subcardinal veins*

_____ **49.** The **stapedius muscle** that moves the stapes ossicle is innervated by

- a) CN V3 b) CN XII c) CN III d) CN VII e) cervical nerves C2 and C 3

_____ **50.** **Pharyngeal groove 1** gives rise to

- a) internal auditory meatus b) external auditory meatus
c) eustachian tube d) cervical sinus e) primary tympanic cavity

_____ **51.** **Failure of choroid fissure to close** results in

- a) congenital detached retina b) congenital aniridia c) congenital aphakia
d) coloboma iridis e) microphthalmos

_____ **52.** **Aqueous humor** is produced by the

- a) choroid plexus b) trabecular meshwork c) ciliary processes
d) vitreous body e) lens vesicle

_____ **53.** Concerning the **development of the urogenital system**, which of the following statements is true?

- a) The labia minora arise embryologically from paramesonephric ducts.
b) The uterine tubes of the adult female are derived embryologically from uterovaginal primordium.
c) The most common cause of testicular feminization syndrome is the presence of abundant androgen receptors.
d) Round ligament of the uterus is formed from the gubernaculums.
e) The X chromosome carries a gene on its short arm that codes for TDF.

_____ **54.** With regard to the development of the **endocrine system**, which of the following statements is false?

- a) the oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.
b) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of

pharyngeal pouches.

c) the thyroid gland is the first endocrine gland to develop in the embryo.

d) the pituitary gland is ectodermal in origin.

e) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.

_____ 55. The **transitional epithelium lining the ureter** is derived from

- a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

_____ 56. Which of the following is not the adult derivative of the **ureteric bud**?

- a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

_____ 57. Which of the following statements about the **optic nerve** is **false**? The **optic nerve**

- a. is a tract of diencephalon. b. is myelinated by oligodendrocyte. c. is invested by meninges.
d. is not capable of regeneration after transaction. e. is developed from the first pharyngeal pouch.

_____ 58. Concerning the **development of the Urinary System**, which statement is **false**?

- a) Urinary system begins to develop before the genital system
b) The metanephric diverticulum is the primordium of the ureter.
c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
d) Differentiation of the nephrons depends upon induction by the collecting tubules.
e) The kidneys attain their definitive adult position toward the end of the fetal period.

_____ 60. Concerning the **development of the urinary bladder and urethra**, which statement is **false**?

- a) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
b) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.
c) In adults, the urachus is represented by the median umbilical ligament.
d) The urinary bladder becomes a pelvic organ during the 9th week of development.
e) The female urethra is derived from endoderm of the urogenital sinus.

_____ 63. During the **development of the kidney systems**, which statement is **true**?

- a) The transitional epithelium lining the urethra is derived from neuroectoderm.
b) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.

- d) Urine production begins shortly after birth. e) At birth, the kidneys have a lobulated appearance.

_____ **64.** During the development of the **female reproductive system**, which statement is **false**?

- a) The urogenital folds remain unfused and form the labia minora.
- b) The cranial portion of the paramesonephric ducts form the uterine tubes.
- c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.
- d) The phallus forms the mons pubis. e) The labioscrotal swellings form the labia majora.

_____ **65.** With regard to the **development of the male reproductive system**, which statement is **false**?

- a) The genotype of an embryo is established at fertilization.
- b) By week 12, male characteristics of the external genitalia are recognizable.
- c) Sertoli cells are derived from mesoderm. d) Seminiferous tubules are formed by week 20.
- e) In epispadias usually the penis is underdeveloped and curved ventrally a condition known as chordee.

_____ **66.** Concerning the **development of the genital system**, one statement is **false**?

- a) Hypospadias is constantly associated with exstrophy of urinary bladder.
- b) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora.
- c) In testicular feminization syndrome, uterus and uterine tubes are generally absent.
- d) Bilateral cryptorchidism results in sterility. e) None of the above.

_____ **68.** With regard to the **development of gonads**, which statement is **false**?

- a) Sex of an embryo is determined genetically at the time of fertilization.
- b) Germ cells do not appear in the genital ridges until the 6th week of development.
- c) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
- d) The surface epithelium of the female gonad develop into oogonia.
- e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ **71.** During the **development of the testes**, which statement is **false**?

- a) A dense tunica albuginea is the characteristic feature of testicular development in the fetus.
- b) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.
- c) The fetal testis produces antimüllerian hormone by sertoli cells at 6-7 weeks
- d) Sertoli cells are derived from the surface epithelium of the testis.
- e) None of the above.

_____ **69.** With regard to the **vestigial structures derived from embryonic genital ducts**, which statement is **false**?

- a) The cranial end of mesonephric duct in males may persist as paradidymis.
- b) The epoophoron in the females correspond to the efferent ductules and duct of epididymis
- c) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- d) The prostatic utricle is homologous to the vagina.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ **70.** Concerning the **development of the genital system**, which statement is **false**?

- a) Inguinal canals develop in both sexes.
- b) Sertoli cells constitute most of the seminiferous epithelium in the fetal testis.
- c) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
- d) Gubernaculum does not play any role in the descent of ovaries.
- e) The persistent processus vaginalis in the female is called canal of nuck.

_____ **72.** With regard to **development of the ovaries**, which statement is **false**?

- a) The cranial part of the gubernaculum becomes the ovarian ligament.
- b) As the ovary separates from the regressing mesonephros, it is suspended by mesovarium.
- c) The surface epithelium becomes separated from the follicles in the cortex by tunica albuginea.
- d) The primordial follicles are derived from the rete ovarii.
- e) No oogonia are formed postnatally.

_____ **73.** With regard to **development of the eye**, which statement is **true**?

- a. Failure of choroid fissure to close results in congenital aphakia.

- b. The epithelium of the ciliary body is derived from endoderm.
- c. The optic nerve is a tract of diencephalon that is completely myelinated during the third trimester.
- d. The hyaloid canal (cloquet's canal) is found in the vitreous body.
- e. Aqueous humor is produced by the choroid plexus.

_____ **74. With regard to development of the ear, which statement is true?**

- a. The spiral organ of Corti is derived from neural crest.
- b. The middle ear cavity develops from the otic vesicle.
- c. The utricular portion of the otic vesicle gives rise to cochlear duct.
- d. Perilymph enters the subarachnoid space via the endolymphatic sac.
- e. None of the above.

_____ **76. Abnormal development of canal of Schlemm or the iridocorneal filtration angle causes**

- a. coloboma iridis
- b. congenital cataract
- c. congenital glaucoma
- d. cyclopia
- e. microphthalmia

_____ **78. With regard to the development of the head and neck,**

- a) sphenoidal sinuses develop during the early fetal life (10 to 11 weeks)
- b) the critical period of palate development is from the end of the 6th week until the beginning of the 9th week.
- c) taste buds form on the dorsal surface of the tongue only.
- d) all parenchymal (secretory) tissue of the salivary glands arises by proliferation of the oral epithelium.
- e) by late fetal period, the nasolacrimal duct drains into the inferior meatus in the lateral wall of the nasal cavity.

_____ **79. Concerning the development of vasculature,**

- a) distal parts of the 3rd pair of aortic arches join the dorsal aortae to form the internal carotid arteries.
- b) the left 4th aortic arch becomes the proximal part of the left subclavian artery.
- c) the distal part of the right 6th aortic arch becomes the ductus arteriosus.
- d) the superior vena cava forms from the right anterior cardinal vein and the right posterior cardinal vein.
- e) after birth, the proximal parts of the umbilical arteries persist as the superior vesical arteries.

_____ 77. Utricular portion of the otic vesicle does not give rise to

- a. utricle
- b. semicircular ducts
- c. vestibular ganglion of CNVII
- d. endolymphatic duct and sac
- e. cochlear duct [organ of Corti]

_____ 80. Development of the **Integumentary system** entails that,

- a) the epidermis is derived from surface ectoderm.
- b) castoff cells from the epidermis mix with secretions of sebaceous glands to form the vernix caseosa.
- c) teeth develop from ectoderm, mesoderm and neural crest cells.
- d) eccrine sweat glands develop from mesenchyme, which is derived from the mesoderm underlying the surface ectoderm.
- e) toenails and fingernails begin to develop at the tips of the digits postnatally.

_____ 81. With regard to the **development of the digestive system**,

- a) the foregut may give rise to the lower respiratory system.
- b) the midgut may give rise to the descending colon and sigmoid colon.
- c) the hindgut derivatives are supplied by inferior mesenteric artery.
- d) during the rotation of the stomach, the lesser curvature moves to the left and the greater curvature moves to the right.
- e) the dorsal pancreatic bud forms the uncinate process and part of the head of the pancreas.

_____ 82. During the **development of the digestive system**,

- a) omphalocele is a persistence of the herniation of abdominal contents into the proximal part of the umbilical cord.
- b) the superior two-thirds of the adult anal canal are derived from proctodeum.
- c) the junction of the epithelium is derived from the ectoderm of the proctodeum.
- d) the inferior one-third of the anal canal is supplied with superior rectal artery.
- e) the anal membrane normally breaks down by the end of the second trimester.

83. Concerning the **development of vasculature**,

- a) distal parts of the 3rd pair of aortic arches join the dorsal aortae to form the internal carotid arteries.
- b) the left 4th aortic arch becomes the proximal part of the left subclavian artery.
- c) the distal part of the right 6th aortic arch becomes the ductus arteriosus.
- d) the superior vena cava forms from the right anterior cardinal vein and the right posterior cardinal vein.
- e) after birth, the proximal parts of the umbilical arteries persist as the superior vesical arteries.

GOOD LUCK

37. Action potentials travel fastest along unmyelinated axons.fff

38. Action potentials normally travel from the axon terminal toward the cell body.fff

39. Unmyelinated axons conduct more slowly than myelinated axons. ttt

40. Unmyelinated axons generally have a smaller diameter than myelinated axons. ttt

41. Meissner's and Pacinian corpuscles are usually served by myelinated axons. Ttt

42. Dendrites are found in gray matter.tttt

43. Axons from receptors for slow, burning pain are typically small and unmyelinated.

44. Myelinated axons do not usually enter gray matter.

45. Dendrites are found in gray matter.

46. Among the features which distinguish axons from cell bodies and dendrites is the absence of ribosomes from axons and axon terminals.

47. For long-axon neurons (those whose axons leave the gray matter location where the cell body resides), the volume of axoplasm will usually exceed the volume of cytoplasm in the cell body.

48. Neurons with large rather than small cell bodies usually give rise to relatively long rather than local axons.

49. White matter in the central nervous system consists of axon tracts with abundant myelin, where neuron cell bodies and dendrites are scarce or absent.

50. A single Schwann cell forms myelin around one and only one axon, while a single oligodendroglial cell forms myelin around several separate axons.

- 51. Myelinated axons found in white matter usually begin in gray matter and terminate in gray matter.
- 52. Rapid axoplasmic transport involves movement along microtubules as the primary mechanism, with ATP from mitochondria supplying the energy.
- 53. After damage to a mixed peripheral nerve, motor axons initially degenerate distal to the site of injury.
- 54. Neuron cell bodies respond to axonal injury with observable changes called chromatolysis.
- 55. Motor axons cannot regenerate after peripheral nerve injury.
- 56. The longest axons may be over a meter in length.
- 57. The thickest axons may be over 50 μm in diameter.

Cardiovascular and respiratory

Color not changed, just the size

A typical peripheral mixed nerve includes all of the following EXCEPT:

- a. motor axons.
- b. connective tissue of the perineurium and/or endoneurium.
- c. Schwann cells.
- d. sensory axons.
- e. astrocytes.

Which of the following cell types forms myelin in the central nervous system?

- a. astroglia
- b. oligodendroglia

- c. microglia
- d. Schwann cells
- e. ependymal cells

Which the of the following cell types lines the central canal of the spinal cord and the ventricular system of the brain?

- a. astroglia
- b. oligodendroglia
- c. microglia
- d. Schwann cells
- e. ependymal cells

Choroid plexus consists of:

- a. dense regular connective tissue.
- b. white matter.
- c. gray matter.
- d. ependymal tissue and associated capillaries.
- e. arachnoid granulations.

Choroid plexus is NOT found:

- a. in the lateral ventricles.
- b. in the third ventricle.
- c. in the fourth ventricle.
- d. in the central canal of the spinal cord.
- e. Choroid plexus is found in ALL of the above locations.

An imbalance between production and drainage of cerebrospinal fluid can cause:

- a. Monro's disease.
- b. astigmatism.
- c. presbyopia.
- d. glaucoma.
- e. hydrocephalus.

Pyramidal cells (neurons whose cell bodies have a characteristic, pyramidal shape) are characteristic of the:

- a. retina.
- b. cerebral cortex.
- c. cerebellar cortex.
- d. ventral horn of spinal cord.
- e. dorsal horn of spinal cord.

The electrical signal of an action potential is conveyed into the interior of each muscle fiber by:

- a. sarcomeres.
- b. transverse tubules.
- c. sarcoplasmic reticulum
- d. Z-lines.
- e. thick filaments.

Z-lines represent attachment sites for:

- a. thick filaments (myosin)
- b. thin filaments (actin)
- c. t-tubules
- d. sarcoplasmic reticulum
- e. neuromuscular junctions

In ordinary connective tissue, the predominant component of ground substance is:

- a. collagen.
- b. elastin.
- c. fibrin.
- d. hyaluronic acid.
- e. water.

Which connective tissue component is characterized by high tensile strength and flexibility?

- a. collagen
- b. elastin
- c. glycosaminoglycan
- d. fibroblasts
- e. lipid

Which type of connective tissue is avascular and further characterized by ground substance in the form of a firm, resilient solid with a high proportion of glycosaminoglycans and with cells isolated in lacunae?

- a. adipose tissue
- b. tendon
- c. cartilage
- d. bone
- e. areolar tissue

Cells which secrete the components of bone matrix are called:

- a. fibroblasts.
- b. chondroblasts.
- c. osteoblasts.

- d. osteoclasts.
- e. osteocytes.

Bone formation without a pre-existing cartilage model is called:

- a. intramembranous bone formation.
- b. extramembraneous bone formation.
- c. endochondral bone formation.
- d. trabecular bone formation.
- e. cancellous bone formation.

The activity of osteoclasts is inhibited, decreasing blood calcium levels, by secretion from the:

- a. thyroid follicles.
- b. adrenal cortex.
- c. parathyroid.
- d. thyroid parafollicular cells (C cells).
- e. neurohypophysis.

In the restructuring of the skeleton, bone remodeling:

- a. follows a cycle in which bone formation alternates with resorption.
- b. occurs exclusively in adult bone.
- c. occurs independently of soft tissue growth and external forces.
- d. only occurs at the epiphyses of long bones.
- e. results in the formation of cement lines.

The term most nearly synonymous to "dense bone" is:

- a. spongy bone.
- b. compact bone.
- c. lamellar bone.
- d. woven bone.
- e. cancellous bone.

In compact bone, osteocytes which are not adjacent to capillaries receive or exchange nutrients and metabolites by means of:

- a. processes contacting other osteocytes along canaliculi.
- b. lymph drainage through central (Haversian) canals.
- c. diffusion through the bony matrix.
- d. fenestrations in medullary vessels.
- e. active transport across the endosteum.

The "white" of the eye is primarily:

- a. elastic tissue.
- b. epithelium.
- c. nervous tissue.
- d. collagen.
- e. adipose.

Substantia propria is the name for which tissue component of the cornea?

- a. epithelial tissue
- b. connective tissue
- c. nervous tissue.
- d. muscle tissue

What type of epithelium forms the outer surface of the cornea?

- a. simple squamousepithelium
- b. simple columnar epithelium.
- c. simple cuboidal epithelium.
- d. keratinized stratified squamous epithelium.
- e. non-keratinized stratified squamous epithelium.

The layer of cuboidal cells which forms the innermost surface of the cornea is called:

- a. corneal epithelium
- b. corneal endothelium
- c. pigmented epithelium
- d. substantial propria.
- e. Decemet's membrane

The iris is composed of several tissue types. Most of the visible portion is:

- a. adipose connective tissue.
- b. loose connective tissue.
- c. dense fibrous connective tissue.
- d. nervous tissue.
- e. epithelial tissue.

Which of the following structures is/are acidophilic?

- a. nucleoli.
- b. heterochromatin.
- c. fat droplets.
- d. ribosomes.
- e. mitochondria.

One of the functions of the Golgi apparatus is to:

- a. synthesize protein.

- b. nucleate polymerization of microtubules.
- c. synthesize lysosomal enzymes.
- d. process and package secretory products and glycocalyx.
- e. synthesize carbohydrates.

The site of assembly of ribosomal subunits is the:

- a. rough endoplasmic reticulum.
- b. smooth endoplasmic reticulum.
- c. golgi apparatus.
- d. spindle apparatus.
- e. nucleolus.

Which of the following best describes the function of the condenser lens of the bright-field light microscope:

- a. illuminates specimen.
- b. magnifies specimen.
- c. focusses light at the level of specimen.
- d. gathers light that has passed through specimen.
- e. projects image to the eyepiece.

Which epithelial cell surface specialization is used to move mucous and particles along the surface?

- a. cilia
- b. stereocilia
- c. desmosomes
- d. microvilli
- e. villi

Non-ciliated cuboidal cells in bronchiole epithelium that are thought to be involved in absorption or secretion are called:

- a. chief cells.
- b. clara cells.
- c. goblet cells.
- d. basal cells.
- e. Type I cells.

The lining of the pleural cavity is:

- a. endothelium.
- b. mesothelium.
- c. pseudostratified columnar epithelium.
- d. type I pneumocytes.
- e. type II pneumocytes.

The inner layer of a blood vessel wall, characterised by a simple squamous endothelium supported by a thin layer of connective tissue, is the:

- a. adventitia.
- b. external elastic lamina.
- c. media.
- d. internal elastic lamina.
- e. intima.

A prominent inner elastic membrane (internal elastic lamina), often appearing in cross section as a wavy or sinuous line, is characteristic of:

- a. muscular arteries.
- b. small veins.
- c. large veins.
- d. venules.
- e. capillaries.

Vasa vasorum are:

- a. blood vessels of the myocardium.
- b. nerves that supply the blood vessels.
- c. nerves of the heart.
- d. blood vessels within the walls of the blood vessels.

e. **blood vessels of the endocardium.**

Which of the following features is a normal component of epicardium but NOT of endocardium?

- a. **adipocytes**
- b. **collagen.**
- c. **elastin**
- d. **fibroblasts**
- e. **simple squamous epithelial tissue**

Continuous endothelium is found in:

- a. endocrine organs.
- b. kidney.
- c. **lung.**
- d. spleen.
- e. intestinal villi.

Lymphatic nodules are NOT found in:

- a. **cortex of lymph nodes.**
- b. **tonsils.**
- c. **spleen.**
- d. **respiratory and gastrointestinal mucosa.**
- e. **thymus.**

Red and white pulp describes tissue of the:

- a. lymphatic nodules.
- b. bone marrow.
- c. thymus.
- d. spleen.
- e. lymph node.

Smooth muscle is most substantial in:

- a. the epicardium of the heart.
- b. the media of elastic arteries.
- c. the media of muscular arteries.
- d. the myocardium of the heart.
- e. arterioles.

The sino-atrial (SA) node, the atrio-ventricular (AV) node, and the Purkinje fibers of the myocardium all consist of specialized:

- a. endothelial cells.
- b. fibroblasts.
- c. nerve cells.
- d. smooth muscle cells.
- e. cardiac muscle cells.

Heart valves normally consist of an endothelial surface covering:

- a. cardiac muscle fibers.
- b. hyaline cartilage.
- c. loose areolar connective tissue.
- d. fibrocollagenous and fibroelastic connective tissue.
- e. adipose connective tissue.

Blood vessels are normally encountered in all of the following EXCEPT:

- a. epicardium.
- b. myocardium.
- c. endocardium.
- d. wall of aorta.
- e. respiratory mucosa.

The germinal center of a lymph nodule is the active site of lymphocyte proliferation.

Red pulp of the spleen is comprised of splenic cords, venous sinuses, and reticular fibers.

The heart is lined by endothelium on the inside and mesothelium on the outside.

Hassall's corpuscles are unique and characteristic feature of the thymus.

Reticular fibers do not form a supporting meshwork for cells in lymph nodes. ffff

pericytes are undifferentiated cells around the perimeter of capillaries

1. Structures called acidophilic are those which normally:
 - a. stain red or pink with hematoxylin.
 - b. stain red or pink with eosin.
 - c. stain blue or purple with hematoxylin.
 - d. stain blue or purple with eosin.
 - e. remain unstained with H&E.
2. Cell structures which bind to basic dyes such as hematoxylin are said to be:
 - a. basophilic.
 - b. acidophilic.
 - c. argyrophilic.
 - d. chromophilic.
 - e. metachromatic.

3. Cell structures which bind to acidic dyes such as eosin are said to be:
 - a. basophilic.
 - b. acidophilic.
 - c. argyrophilic.
 - d. chromophilic.
 - e. metachromatic.

4. Structures called basophilic are those which normally:
 - a. stain red or pink with hematoxylin.
 - b. stain red or pink with eosin.
 - c. stain blue or purple with hematoxylin.
 - d. stain blue or purple with eosin.
 - e. remain unstained with H&E.

5. Which of the following structures is/are acidophilic?
 - a. ribosomes.
 - b. heterochromatin.
 - c. fat droplets.
 - d. collagen.
 - e. nucleoli.

6. Which of the following structures is/are acidophilic?
 - a. muscle protein.
 - b. heterochromatin.
 - c. fat droplets.
 - d. ribosomes.
 - e. nucleoli.

7. Which of the following structures is/are acidophilic?
 - a. nucleoli.
 - b. heterochromatin.
 - c. fat droplets.
 - d. ribosomes.
 - e. mitochondria.

8. Which of the following structures is/are basophilic?
 - a. muscle protein.
 - b. collagen.
 - c. Golgi bodies.
 - d. ribosomes.
 - e. mitochondria.

9. Which of the following structures is/are basophilic?
 - a. muscle protein.

- b. fat droplets.
- c. collagen.
- d. mitochondria.
- e. nucleoli.

10. Which of the following structures is/are basophilic?

- a. heterochromatin.
- b. fat droplets.
- c. collagen.
- d. mitochondria.
- e. muscle protein.

11. Which of the following structures is/are normally unstained with H&E?

- a. heterochromatin.
- b. fat droplets.
- c. collagen.
- d. mitochondria.
- e. muscle protein.

12. Which of the following structures is/are normally unstained with H&E?

- a. muscle protein.
- b. collagen.
- c. Golgi bodies.
- d. ribosomes.
- e. mitochondria.

13. Which of the following structures is/are normally unstained with H&E?

- a. heterochromatin.
- b. ribosomes.
- c. collagen.
- d. mitochondria.
- e. connective tissue ground substance.

14. A cell specialized for synthesizing protein can be recognized by cytoplasm containing especially large numbers of which of the following organelles?

- a. mitochondria.
- b. microtubules.
- c. lysosomes.
- d. ribosomes.
- e. centrioles.

15. On histological slides stained with hematoxylin and eosin (H&E), the color of cytoplasm in cells specialized for protein synthesis is:

- a. red or pink.

- b. blue or purple.
- c. yellow, brown, or black.
- d. greenish gray.
- e. pale (unstained).

16. A cell which produces an exceptionally abundant amount of ATP can be recognized by cytoplasm containing especially large numbers of which of the following organelles?

- a. mitochondria.
- b. microtubules.
- c. lysosomes.
- d. ribosomes.
- e. centrioles.

17. On histological slides stained with hematoxylin and eosin (H&E), the color of cytoplasm in cells containing many mitochondria is:

- a. red or pink.
- b. blue or purple.
- c. yellow, brown, or black.
- d. greenish gray.
- e. pale (unstained).

18. 1000 micrometers is equivalent to:

- a. 1 centimeter.
- b. 100 millimeters.
- c. 10 millimeters.
- d. 1 millimeter.
- e. 0.1 millimeter.

19. The diameter of a typical red blood cell is about:

- a. 0.6 micrometer.
- b. 6 micrometers.
- c. 60 micrometers.
- d. 600 micrometers.
- e. 6000 micrometers.

20. The diameter of a typical mitochondrion is about:

- a. 0.5 micrometer.
- b. 5 micrometers.
- c. 50 micrometers.
- d. 500 micrometers.
- e. 5000 micrometers.

21. The diameter of a typical adipocyte is about:

- a. 0.5 micrometer.

- b. 5 micrometers.
- c. 50 micrometers.
- d. 500 micrometers.
- e. 5000 micrometers.

22. A brown pigment comprising the indigestible waste products of cell metabolism is called:

- a. melanin.
- b. lipofuchsin.
- c. hematoxylin.
- d. hemoglobin.
- e. eosin.

23. During mitosis, chromosomes are aligned in the middle of the spindle during:

- a. prophase.
- b. metaphase.
- c. anaphase.
- d. telophase.
- e. interphase.

24. The mitotic spindle consists of:

- a. chromosomes.
- b. histones.
- c. microtubules.
- d. ribosomes.
- e. lipid.

25. One of the functions of the Golgi apparatus is to:

- a. synthesize protein.
- b. nucleate polymerization of microtubules.
- c. synthesize lysosomal enzymes.
- d. process and package secretory products and glycocalyx.
- e. synthesize carbohydrates.

26. The organelle associated with energy (ATP) production is the:

- a. chromosome.
- b. centriole.
- c. ribosome.
- d. mitochondrion.
- e. endoplasmic reticulum.

27. Protein molecules are assembled on:

- a. chromosomes.
- b. histones.
- c. microtubules.

- d. ribosomes.
- e. lipid.

28. The site of assembly of ribosomal subunits is the:

- a. rough endoplasmic reticulum.
- b. smooth endoplasmic reticulum.
- c. golgi apparatus.
- d. spindle apparatus.
- e. nucleolus.

29. The stain of choice to demonstrate glycogen and/or mucus in cells is:

- a. periodic acid Schiff (PAS).
- b. hematoxylin.
- c. eosin.
- d. lipofuchsin.
- e. hemosiderin.

30. Which of the following best describes the function of the condenser lens of the bright-field light microscope:

- a. illuminates specimen.
- b. magnifies specimen.
- c. focusses light at the level of specimen.
- d. gathers light that has passed through specimen.
- e. projects image to the eyepiece.

31. Which region of the nucleus appears only lightly stained with routine histological procedures and is the site of mRNA synthesis?

- a. nucleolus.
- b. euchromatin.
- c. heterochromatin.
- d. centrosome.
- e. ribosome.

32. Which of the following blood cells develops into a macrophage in loose connective tissue?

- a. neutrophil
- b. eosinophil
- c. basophil
- d. monocyte
- e. lymphocyte

33. Phagocytosis is exhibited by:

- a. fibroblasts.
- b. adipocytes.
- c. mast cells.

- d. macrophages.
- e. basophils.

34. Which of the following cells is NOT normally found in loose connective tissue?

- a. mast cell.
- b. neutrophil.
- c. lymphocyte.
- d. adipocyte.
- e. macrophage.

35. Elastin is synthesized by:

- a. fibroblasts.
- b. mast cells.
- c. lymphocytes.
- d. keratinocytes.
- e. plasma cells.

36. Collagen is synthesized by:

- a. fibroblasts.
- b. mast cells.
- c. lymphocytes.
- d. keratinocytes.
- e. plasma cells.

37. The cell responsible for the synthesis, maintenance, and turnover of extracellular matrix is the:

- a. fibroblast.
- b. mast cell.
- c. lymphocyte.
- d. keratinocyte.
- e. plasma cell.

38. Which of the following proteins is assembled into extracellular fibers?

- a. collagen.
- b. keratin.
- c. actin.
- d. myosin.
- e. tropomyosin.

39. The predominant matrix component of ordinary loose connective tissue is:

- a. collagen.
- b. elastin.
- c. GAG (glycosaminoglycan).
- d. water.

40. Which of the following is a function for brown (multilocular) fat but NOT also a function of white (unilocular) fat?
- a. lipid storage.
 - b. thermal insulation.
 - c. heat generation.
 - d. cushioning (shock absorption).
41. Which of the following is a permanent resident cell (as opposed to a transient immigrant) of connective tissue?
- a. neutrophil.
 - b. eosinophil.
 - c. plasma cell.
 - d. fibroblast.
 - e. monocyte.
42. Which of the following secretes histamine and other chemical mediators in the vicinity of small vessels?
- a. mast cell.
 - b. lymphocyte.
 - c. fibroblast.
 - d. monocyte.
 - e. adipocyte.
43. Plasma cells are derived from:
- a. mast cells.
 - b. B-lymphocytes.
 - c. fibroblasts.
 - d. monocytes.
 - e. adipocytes.
44. Which of the following is a transient cell (as opposed to a permanent resident) of connective tissue?
- a. fibroblast.
 - b. mast cell.
 - c. mesenchymal cell.
 - d. lymphocyte.
 - e. adipocyte.
45. All epithelia rest on a thin layer called:
- a. submucosa.
 - b. lamina propria.
 - c. dermis.
 - d. basement membrane.
 - e. serosa.

46. An epithelium which consists of a single layer of cells is designated as:
- stratified.
 - simple.
 - columnar.
 - pseudostratified.
 - transitional.
47. An epithelium which consists of more than one layer of cells is designated as:
- stratified.
 - simple.
 - columnar.
 - pseudostratified.
 - transitional.
48. An epithelium which appears to be composed of more than one layer of cells but which really consists of a single layer is designated as:
- stratified.
 - simple.
 - columnar.
 - pseudostratified.
 - transitional.
49. An epithelium whose surface cells are flattened is called:
- squamous.
 - cuboidal.
 - columnar.
50. An epithelium whose surface cells are much taller than wide is called:
- squamous.
 - cuboidal.
 - columnar.
51. An epithelium whose surface cells are about as tall as wide is called:
- squamous.
 - cuboidal.
 - columnar.
52. The portion of an epithelial cell which is closest to the basement membrane is called the cell's:
- basal surface.
 - lateral surface.
 - apical surface.
53. The portion of an epithelial cell which is attached to adjacent epithelial cells is called the cell's:

- a. basal surface.
- b. lateral surface.
- c. apical surface.

54. The portion of an epithelial cell which faces the outer, free surface of the epithelium is called the cell's:

- a. basal surface.
- b. lateral surface.
- c. apical surface.

55. Stratified squamous epithelium lines the:

- a. esophagus.
- b. trachea.
- c. stomach.
- d. small intestine.
- e. urinary bladder.

56. *Mesothelium is:*

- a. ***simple squamous epithelial tissue.***
- b. *stratified squamous epithelial tissue.*
- c. *simple cuboidal epithelial tissue.*
- d. *stratified cuboidal epithelial tissue.*
- e. *simple columnar epithelial tissue*

57. *Endothelium is:*

- a. ***simple squamous epithelial tissue.***
- b. *stratified squamous epithelial tissue.*
- c. *simple cuboidal epithelial tissue.*
- d. *stratified cuboidal epithelial tissue.*
- e. *simple columnar epithelial tissue*

58. Endothelium lines all of the following EXCEPT:

- a. arteries.**
- b. veins.**
- c. capillaries.**
- d. lymphatic vessels.**
- e. secretory ducts.**

59. Mesothelium lines:

- a. the respiratory tract.**
- b. the urinary tract.**
- c. the peritoneal, pleural and pericardial cavities.**
- d. the nasal cavity.**
- e. the oral cavity.**

60. Which type cell junction creates a site of low electrical resistance between adjacent cells?

- a. tight junction / zonula occludens.
- b. gap junction.
- c. adhering junction (desmosome).

61. Which type cell junction provides a relatively strong attachment between adjacent cells?

- a. tight junction / zonula occludens.
- b. gap junction.
- c. adhering junction / macula adherens / desmosome.

62. Which cell junction establishes a barrier to diffusion between adjacent epithelial cells?

- a. tight junction / zonula occludens.
- b. gap junction.
- c. adhering junction / macula adherens / desmosome.

63. The epithelial cell specialization which vastly increases the absorptive surface area of the cell is:
- cilia.
 - stereocilia.
 - desmosomes.
 - microvilli.
 - keratinization of a stratified epithelium.
64. Which epithelial cell surface specialization is used to move mucous and particles along the surface?
- cilia.
 - stereocilia.
 - desmosomes.
 - microvilli.
 - villi.
65. Which epithelial cell surface modification is best suited to resist abrasive wear and tear?
- cilia.
 - stereocilia.
 - desmosomes.
 - microvilli.
 - keratinization of a stratified epithelium.
66. Which surface is NOT lined by epithelial tissue?
- surface of skin.
 - cornea surface of eye.
 - luminal surface of intestinal tract.
 - luminal surface of trachea and alveoli of lungs.
 - luminal surface of urinary tract.
 - surfaces of bones and muscles.
67. The largest hemopoietic cell in bone marrow is the:
- stem cell.
 - erythroblast.
 - myelocyte.
 - metamyelocyte.
 - megakaryocyte.
68. Which is (normally) the most common white blood in circulating blood?
- lymphocyte.
 - monocyte.
 - neutrophil.
 - eosinophil.
 - basophil..
69. Which is (normally) the SECOND most common white blood in circulating blood?

- a. lymphocyte.
- b. monocyte.
- c. neutrophil.
- d. eosinophil.
- e. basophil.

70. Which of the following cell types immigrates first (i.e., most quickly) from blood into connective tissue, during the early stages of an inflammatory response?

- a. lymphocyte.
- b. monocyte.
- c. neutrophil.
- d. eosinophil.
- e. basophil..

71. Circulating antibodies are produced by:

- a. B-lymphocytes.
- b. monocytes.
- c. neutrophils.
- d. eosinophils.
- e. basophils.

72. What is the component of peripheral blood provided by megakaryocytes?

- a. erythrocytes.
- b. neutrophils.
- c. basophils.
- d. platelets.
- e. plasma.

73. Which is the smallest white blood cell found in peripheral blood?

- a. lymphocyte.
- b. monocyte.
- c. neutrophil.
- d. eosinophil.
- e. basophil.

74. Which blood cell has a nucleus pinched into several lobes connected by narrow constrictions?

- a. erythrocyte.
- b. neutrophil.
- c. lymphocyte.
- d. platelet.
- e. monocyte.

75. Which component of blood consists of a small bit of cytoplasm without a nucleus?

- a. basophil.

- b. neutrophil.
- c. lymphocyte.
- d. platelet.
- e. monocyte.

76. Which is a large white blood cell that represents about 3 to 8% of circulating leukocytes?

- a. lymphocyte.
- b. monocyte.
- c. neutrophil.
- d. eosinophil.
- e. basophil.

77. Which of the following white blood cells contains specific granules that have little affinity for dyes, are similar to lysosomes, and are used to digest bacteria?

- a. lymphocyte.
- b. platelet.
- c. neutrophil.
- d. eosinophil.
- e. basophil.

78. Which of the following white blood cells is associated with allergies and parasite infections and contain specific granules that have an affinity for eosin and other acidic dyes?

- a. lymphocyte.
- b. platelet.
- c. neutrophil.
- d. eosinophil.
- e. basophil.

79. Which of the following white blood cells is most similar to a circulating mast cell (and is the least common of the circulating leukocytes)?

- a. lymphocyte.
- b. platelet.
- c. neutrophil.
- d. eosinophil.
- e. basophil.

80. Which of the following is the name given to an immature red blood cell which still contains remnants of basophilic cytoplasm?

- a. erythrocyte.
- b. platelet.
- c. reticulocyte.
- d. megakaryocyte.
- e. metamyelocyte.

81. Hemopoiesis is an important function for:
- tonsils.
 - skin.
 - cartilage.
 - bone marrow.
 - cardiac muscle.
82. Which of the following may undergo further cell division after leaving the hemopoietic organ of its origin?
- lymphocyte.
 - erythrocyte.
 - neutrophil.
 - eosinophil.
 - basophil.
83. Which of the following blood cell types will normally re-enter the bloodstream after leaving circulation and immigrating into peripheral tissues?
- lymphocyte.
 - monocyte.
 - neutrophil.
 - eosinophil.
 - basophil.
84. The term "polymorphonuclear leukocytes" includes:
- neutrophils and eosinophils.
 - basophils, eosinophils, and lymphocytes.
 - lymphocytes and monocytes.
 - T-cells, B-cells and neutrophils.
85. Reticulocytes are:
- part of the reticulo-endothelial system.
 - immature RBCs.
 - macrophages.
 - dendritic cells.
 - none of the above
86. A macrophages can originate from a:
- lymphocytes.
 - monocytes.
 - neutrophils.
 - eosinophils.
 - basophils.
87. Which type of muscle is syncytial, with many nuclei per "cell"?

- a. skeletal muscle.
- b. cardiac muscle.
- c. smooth muscle.

88. Which type of muscle has the largest fibers?

- a. skeletal muscle.
- b. cardiac muscle.
- c. smooth muscle.

89. Which type of muscle consists of individual cells attached end-to-end by intercalated disks?

- a. skeletal muscle.
- b. cardiac muscle.
- c. smooth muscle.

90. Which type of muscle consists of individual cells that may be several hundred micrometers in length but only a few micrometers in diameter?

- a. skeletal muscle.
- b. cardiac muscle.
- c. smooth muscle.

91. Which type of muscle has its nuclei located at the periphery of its muscle fibers, close to the fiber membrane?

- a. skeletal muscle.
- b. cardiac muscle.
- c. smooth muscle.

92. The predominant connective tissue element in the dermis is:

- a. lymphocytes.
- b. mast cells.
- c. macrophages.
- d. elastin.
- e. collagen.

93. The tissue composition of hair follicles is most similar to:

- a. epidermis.
- b. hypodermis.
- c. sweat gland.
- d. papillary layer of dermis.
- e. reticular layer of dermis.

94. Arrector pili consist of:

- a. glandular epithelium.
- b. columnar epithelium.

- c. squamous epithelium.
- d. lymphocytes.
- e. smooth muscle.

95. The glands most closely associated with hair follicles are:

- a. eccrine sweat glands.
- b. apocrine sweat glands.
- c. sebaceous glands.
- d. lymphatic glands.
- e. submucosal glands.

EMBRYOLOGY

_____ 1. Which one of the following structures changes its position because of the **head fold**?

- a) Oropharyngeal membrane
- b) Septum transversum
- c) Primitive heart
- d) a & b
- e) All of the above

_____ 2. Most **birth defects** occur during:

- a) The first 2 weeks
- b) The embryonic period
- c) The fetal period
- d) Early childhood
- e) None of the above

_____ 3. The **cranial half of the 1st cervical sclerotome** gives rise to:

- a) Upper part of body of the 1st cervical vertebra.
- b) Lower part of body of the 1st cervical vertebra.
- c) Whole body of the 1st cervical vertebra.
- d) Part of base of occipital bone.
- e) Part of roof of the cranial cavity

_____ 4. Which of the following **muscles** develop from the epimeres?

- a) Intercostal muscles.
- b) The strap muscles of the neck.
- c) Rectus abdominis.
- d) Quadratus lumborum.
- e) Extensor muscles of the back.

_____ 5. Failure of **fusion of the neural arches** of the developing vertebrae could lead to:

- a) Achondroplasia
- b) Acromegally
- c) Spina bifida
- d) Brachycephaly
- e) Amelia

_____ 6. Which of the following gives rise to most muscles of the tongue?

- a) Lateral mesoderm
- b) Local splanchnic mesenchyme
- c) Myotomes of the occipital somites
- d) All of the above
- e) None of the above

_____ 7. Frontonasal prominence gives rise to the:

- a) Forehead
- b) Sides of the face
- c) Sides of the nose
- d) All of the above
- e) None of the above

_____ 8. Which of the following is **not** a component of vernix caseosa?

- a) Secretions from sebaceous gland
- b) Secretions of apocrine sweat glands
- c) Fallen epidermal cells
- d) Fallen hairs
- e) None of the above

_____ 9. Structures derived from endoderm of the second pharyngeal pouch include:

- a) Thymus gland b) Palatine tonsil c) Thyroid gland
- d) Parafollicular cells of thyroid gland e) All of the above

_____ 10. Germ layer that does **not** give rise to the tympanic membrane

- a) Ectoderm b) Mesoderm c) Endoderm
- d) b & c e) None of the above

_____ 11. The muscular part of interventricular septum is formed by:

- a) Downward extension of the spiral aortopulmonary septum
- b) Fusion of superior and inferior endocardial cushions of the ventricular wall
- c) Fusion of right and left endocardial cushions of the ventricular wall
- d) Upward growth of a midline septum from the floor of a common ventricle
- e) Expansion of the lateral aspects of primitive ventricle with the midline region remained unexpanded

_____ 12. The common carotid arteries develop from:

- a) Truncus arteriosus b) 1st aortic arch arteries c) 2nd aortic arch arteries
- d) 3rd aortic arch arteries e) Dorsal aortae

_____ 13. In fetal blood circulation oxygenated blood is not contaminated by deoxygenated blood in which of the following sites?

- a) Left atrium b) Right atrium c) Inferior vena cava
- d) Descending aorta e) None of the above

_____ 14. Which of the following veins participates in the formation of the inferior vena cava?

- a) Right vitelline vein b) Right posterior cardinal vein
- c) Right umbilical vein d) Left umbilical vein e) Right posterior cardinal vein

_____ 15. Portal vein is derivative of:

- a) Right vitelline vein
- b) Left vitelline vein
- c) Anastomosis between right umbilical and right vitelline veins
- d) Anastomosis between left and right vitelline veins
- e) Anastomosis between left umbilical and left vitelline veins

_____ 16. After birth umbilical vein becomes:

- a) Medial umbilical ligament b) Median umbilical ligament
- c) Lateral umbilical ligament d) Ligamentum teres hepatis
- e) Ligamentum venosum

_____ 17. Which of the following cells of the lung develop from mesoderm?

- a) Type I Pneumocytes. b) Type II Pneumocytes. c) Alveolar dust cells.
- d) Clara cells e) Goblet cells

_____ 18. Which of the following statements is correct regarding to the development of the gut?

- a) Esophagus was short during early development, but gets elongated towards the end of embryonic period.
- b) The midgut was connected to the yolk sac by the allantois.

- c) The primitive gut tube is in open communication with the amniotic cavity.
- d) The site of junction of foregut and midgut derivatives is the 2nd part of duodenum just proximal to the entrance of common bile duct.
- e) All are correct

_____ 19. Concerning the development of pancreas

- a) The ventral bud arises from the proximal part of haptic bud.
- _____ ventral bud rotates & comes to lie anterior to the dorsal bud.
- c) The ventral bud arises from the distal part of foregut, while the dorsal bud arises from the proximal part of midgut.
- _____ exocrine portion develops from endoderm, while the endocrine portion develops from splanchnic mesoderm.
- _____ ventral bud appears before the appearance of the dorsal bud.

_____ 20. Which of the following statements is **not correct** regarding to the development of the coelomic cavities?

- a) Peritoneal cavity was temporarily divided into two by the gut and its primitive mesenteries.
- b) Develop within the intermediate mesoderm.
- c) Are lined by both splanchnic and somatic mesoderm layers.
- d) Primitive pericardial cavity gives rise to the serous pericardium.
- e) Pericardioperitoneal canals are invaginated by the growing lungs.

_____ 21. The crura of the diaphragm develops from:

- a) Septum transversum.
- b) Dorsal mesentery of the esophagus.
- c) Pleuroperitoneal membrane.
- d) Myotomes of the body wall.
- e) None of the above.

_____ 22. Which one of the following is **not** a derivative of the neural crest?

- a) Adrenal chromaffin cells
- b) Melanocytes
- c) Odontoblasts
- d) Enteric nervous system
- e) None of the above

_____ 23. Which of the following nuclei is **not** developed from the alar plate of the neural tube?

- a) Olivary nucleus
- b) Cuneate nucleus
- c) Thalamus
- d) Superior colliculus
- e) None of the above

_____ 24. Primary brain vesicle that remains relatively unchanged as compared to the others is:

- a) Rhombencephalon
- b) Prosencephalon
- c) Telencephalon
- d) Mesencephalon
- e) Diencephalons

_____ 25. During the development of the central nervous system zone of the wall of the neural tube to be occupied by neuroblasts forming the gray matter is:

- a) Marginal zone
- b) Ventricular zone
- c) Mantle zone
- d) Ependymal zone
- e) Neuroepithelial zone